Abdulqader Alshihabi

Software Engineer | Backend & AI Engineering

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SUMMARY

Computer Science student graduating in January 2027 (GPA: 4.62 / 5.00), with practical experience developing scalable backend architectures, full-stack enterprise platforms, and production-grade deep learning vision systems. Proficient in Python, PyTorch, Django, React, TypeScript, and PostgreSQL, with proven contributions across medical diagnostic AI, edge computer vision, semantic vector search, and enterprise ERP workflows.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, Java, Go

Backend & Cloud: Django, Django REST Framework, FastAPI, RESTful APIs, Role-Based Access Control (RBAC), Supabase, Docker, Azure

Frontend Development: React, Next.js, TypeScript, HTML5, CSS3, Tailwind CSS, Canvas API

Databases & Vector Data: PostgreSQL, pgvector, PL/pgSQL, Relational Schema Design, Query Optimization, Vector Embeddings

AI, Machine Learning & Vision: PyTorch, EfficientNet, YOLO11, Faster R-CNN, OpenAI CLIP, ONNX Runtime Web, Computer Vision, Scikit-learn, LLMs

Core Methodologies & Tools: Git, GitHub Actions (CI/CD), Unit Testing, SDLC, Agile/Scrum, Linux, Postman, WebSockets

WORK EXPERIENCE

Research & Innovation Consultant (Part-time) **Jul 2026 – Present**

Institute of Research and Consulting Studies, Islamic University of Madinah | Madinah, Saudi Arabia

- Engineered a locally hosted LLM extraction pipeline using Python, Ollama, and Qwen, processing 25,000+ unstructured academic research papers from the university repository, OpenAlex, and Scopus with zero cloud API costs.
- Accelerated comprehensive research data extraction from an estimated several months of manual review to just 10 days, implementing data validation, Arabic text normalization, and retry mechanisms to improve extraction reliability.
- Organized researcher profiles and domain expertise by generating dense vector representations with multilingual Sentence-Transformers and clustering techniques, designing optimized PostgreSQL schemas for relational and vector queries.
- Collaborated with the platform engineer to deliver research analytics for the commercialization engine and prepared technical progress reports and milestone presentations for university leadership.

Software Engineer Intern **Jun 2026 – Jul 2026**

Asaqefah for Business Development | Saaed HR & Payroll ERP | Madinah, Saudi Arabia

- Contributed to Saaed, an enterprise HR and Payroll ERP serving 30+ employees, developed to replace disconnected departmental spreadsheets with integrated transactional models.
- Developed core backend services with Django REST Framework and interactive frontend views with React and TypeScript on a PostgreSQL database.
- Accelerated payroll approval and finance reconciliation workflows by over 4x by automating the transfer of validated payroll calculations directly into Finance general ledgers.
- Implemented role-based access control (RBAC) and strict server-side validation rules to secure sensitive employee records and maintain transactional consistency across cross-departmental workflows.

PROJECTS

Bone Fracture Diagnostic Classification System **Mar 2025**

github.com/Abdulqader-Alshihabi/Bone-Fracture-Classifier | *Technologies: PyTorch, EfficientNet-B0, ONNX Runtime Web, Scikit-learn, Medical Imaging, Canvas API*

- Trained and evaluated a deep convolutional neural network (EfficientNet-B0) on 428 clinical radiographs to diagnose 12 distinct orthopedic fracture types with 18ms inference latency.
- Engineered an interactive browser PACS radiograph studio featuring contrast inversion, window-level adjustment, and automated top-3 differential diagnosis probability reporting.

SiteSight – Edge AI Construction Safety & PPE Computer Vision **May 2025**

github.com/Abdulqader-Alshihabi/SiteSight-Safety-Vision | *Technologies: YOLO11, PyTorch, ONNX Runtime Web, JavaScript, Canvas API, WebAssembly*

- Engineered an in-browser edge computer vision system running real-time YOLO11 inference (1x3x640x640 input tensor) to detect PPE safety compliance without server GPU infrastructure.
- Implemented client-side letterbox padding, multi-class confidence filtering, and non-maximum suppression (NMS) over 8,400 predictions to log hardhat and vest compliance directly on HTML5 Canvas.

Faster R-CNN Deep Object Detection Pipeline **Aug 2025**

github.com/Abdulqader-Alshihabi/Faster-RCNN-Object-Detection | *Technologies: PyTorch, ResNet-50 FPN, TorchVision, ONNX, FastAPI, PASCAL VOC, JavaScript*

- Built and trained a two-stage Faster R-CNN detector with a ResNet-50 Feature Pyramid Network (FPN) backbone across 20 object classes on 4,952 test images, achieving 24ms inference latency.
- Engineered multi-file ONNX weight graph chunking for client-side model execution and deployed a companion FastAPI prediction service with automated validation suites.

Islamic University CRIS – Research Information System **Oct 2025**

github.com/Abdulqader-Alshihabi/Islamic-University-CRIS | *Technologies: React, TypeScript, PostgreSQL, PL/pgSQL, Supabase, Scopus API, OpenAlex API*

- Centralized and reconciled 7,000+ academic publications and 70,000+ verified citations across Scopus and OpenAlex APIs, resolving fragmented institutional research records.
- Designed automated ETL pipelines in PostgreSQL and PL/pgSQL with multi-field deduplication algorithms, improving the speed and accuracy of faculty post-publication award evaluations.

res-handler – AI Semantic Search & Research Matcher **Sep 2025**

github.com/Abdulqader-Alshihabi/res-handler | *Technologies: Python, Go, PostgreSQL, pgvector, Azure OpenAI Embeddings, Docker*

- Engineered an AI semantic vector search service converting problem queries into 1,536-dimensional embeddings, indexing 5,000+ academic papers to match faculty expertise.
- Implemented cosine distance similarity lookup with pgvector over dense representations, achieving sub-200ms query retrieval latency across high-dimensional embeddings.

EDUCATION

Bachelor of Science in Computer Science **Expected Graduation: January 2027 | GPA: 4.62 / 5.00**

Islamic University of Madinah | Madinah, Saudi Arabia

Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Engineering, Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Operating Systems, Computer Networks

CERTIFICATIONS

- **Advanced Artificial Intelligence (Computer Vision) –** KAUST Academy & Universidad Politécnica de Madrid (2026)
- **Introduction to Artificial Intelligence (with Distinction) –** KAUST Academy & Universidad Politécnica de Madrid (2026)
- **KAUST Academy AI Specialization Selection –** KAUST Academy & Islamic University of Madinah (2026)
- **Asaqefah SandBox Innovation Program –** Asaqefah & Amazon Web Services (AWS) (2026)
- **Introductory Artificial Intelligence Program –** KAUST Academy & NTDP (2025)
- **Improving Deep Neural Networks: Hyperparameter Tuning, Regularization & Optimization –** DeepLearning.AI (2024)
- **Mathematics for Machine Learning: Calculus, Linear Algebra, Probability & Statistics –** DeepLearning.AI (2024)